

Claim Classifier | TikTok Data Team

Executive Summary



Overview:

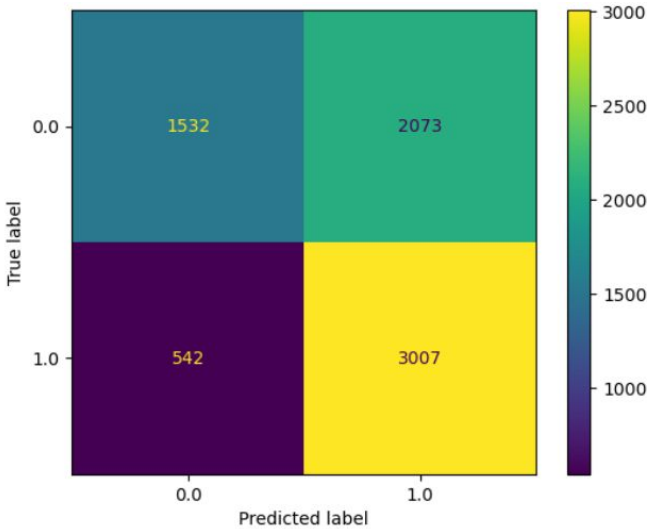
As part of the project of developing a machine learning model to classify claim videos. Our team developed a logistic regression model for predicting author verification status which seemed to have high effect on claims. This reports shows the results and the evaluation of the model and some important insights gained during the process.

Key Insights

- At first EDA were done to clear the data from outliers and missing values that could affect the model. Also for features engineering of variables.
- The model showed good, yet not ideal results.
- Verification status appeared to be correlated with the independent variables (hosen based on variables that obey model's assumptions) with parameters shown below:

	Beta
video_duration_sec	0.001343
video_view_count	-0.000002
video_share_count	0.000004
video_download_count	-0.000175
video_comment_count	0.000340
Length	0.003594
claim_status_opinion	0.000062
author_ban_status_banned	-0.000002
author_ban_status_under review	-0.000001

Details



Alt text: Figure shows the confusion matrix of the model.

Precision: 0.59
F1-Score: 0.7
Recall: 0.85
Accuracy: 0.63

Key Metrics of The Model

Next Steps

- It is recommended to exclude variables that are not highly correlated with verification status and trying to develop the model by minimizing false positive errors.
- Once the model has achieved its most effective state, we can move on into putting it into use for the development of more advanced machine learning model as its predictions will be of high value.